

Data Warehouse MCQs - Answer Key

1. What is the main purpose of a Data Warehouse?

- A) Real-time transaction processing
- B) Storing raw, unstructured data
- C) Analytical processing and decision support ✓**
- D) Replacing operational databases

2. Which system is optimized for fast updates and transactional integrity?

- A) OLAP
- B) Data Warehouse
- C) OLTP ✓**
- D) Data Mart

3. What is the main characteristic of OLAP systems?

- A) High-speed transactions
- B) Complex analytical queries ✓**
- C) Real-time data updates
- D) Only stores structured data

4. Which type of modeling is commonly used in a Data Warehouse?

- A) ER Modeling
- B) Dimensional Modeling ✓**
- C) Relational Modeling
- D) Hierarchical Modeling

5. What is a Data Lake used for?

- A) Storing only structured data
- B) Processing online transactions
- C) Storing large volumes of raw and unstructured data ✓**
- D) Replacing a data warehouse

6. A Data Warehouse is mainly used for:

- A) Operational reporting
- B) Transactional processing
- C) Analytical processing ✓**
- D) Storing unstructured data

7. A Data Mart is:

- A) A subset of a data warehouse ✓**
- B) A replacement for a data warehouse
- C) Used for transactional processing
- D) Only stores structured data

8. In a Star Schema, how are dimension tables connected to the fact table?

- A) One-to-One relationship ✗ (Correct Answer: C) One-to-Many relationship)**

9. The Snowflake Schema is different from Star Schema because:

- A) It has denormalized dimension tables
- B) It has normalized dimension tables** ✓
- C) It does not support queries
- D) It does not store historical data

10. Which component is part of an advanced Data Warehouse architecture?

- A) Only relational databases
- B) Streaming and real-time data integration** ✓
- C) NoSQL data storage
- D) Both B and C

11. In a Data Warehouse, data is typically updated:

- A) Every second
- B) Periodically through batch processing** ✗ (You chose C, correct answer is B)
- C) In real time like an OLTP system
- D) Never

12. What is the purpose of ETL in a Data Warehouse?

- A) Extracting, Transforming, and Loading data** ✓
- B) Encrypting, Testing, and Logging
- C) Evaluating, Tuning, and Learning
- D) Executing, Tracking, and Linking

13. Which schema is best for complex queries and reporting?

- A) Star Schema
- B) Snowflake Schema** ✗ (Correct Answer: C) Both A and B)

14. Which component in Data Warehouse ensures data quality and consistency?

- A) ETL Process** ✓
- B) OLTP System
- C) NoSQL Database
- D) Load Balancer

15. A Fact Table primarily stores:

- A) Transactional data
- B) Dimension attributes
- C) Aggregated metrics and measures** ✗ (Correct Answer: D) Both A and C)

16. Which of the following is a benefit of a Data Warehouse?

- A) Faster transaction processing
- B) Better decision-making through analytics** ✓
- C) Reduces the need for databases
- D) Real-time updates

17. In OLAP, what does "drill-down" mean?

- A) Aggregating data into higher-level summaries
- B) Moving from summarized data to detailed data** ✓
- C) Removing unnecessary data
- D) Extracting data from different sources

18. What is SCD in a Data Warehouse?

- A) Slowly Changing Dimension** ✓
- B) System Control Data
- C) Structured Columnar Database
- D) Sequential Column Details

19. What does a Surrogate Key represent in a Data Warehouse?

- A) A primary key from a source system
- B) A unique, system-generated key** ✓
- C) A foreign key from another table
- D) A user-defined key

20. Which approach is commonly used for integrating Big Data with a Data Warehouse?

- A) ETL Only
- B) Data Lakes and Real-time Processing** ✓
- C) OLTP systems
- D) No integration needed

Informatica MCQs - Answer Key

1. What is Informatica PowerCenter mainly used for?

- A) Database administration
- B) Real-time transaction processing
- C) Extract, Transform, Load (ETL) processes** ✓
- D) Web application development

2. What is the purpose of an Expression Transformation in Informatica?

- A) Filtering records
- B) Performing row-wise calculations** ✓
- C) Joining multiple sources
- D) Aggregating data

3. What does a Router Transformation do?

- A) Filters out specific data
- B) Splits data into multiple groups based on conditions** ✓
- C) Joins multiple sources
- D) Aggregates data

4. How does a Filter Transformation differ from a Router Transformation?

- A) Filter Transformation sends data to multiple targets
- B) Router Transformation applies a single filter condition

C) Filter Transformation discards rows that don't meet the condition ✓

D) Router Transformation cannot send data to multiple targets

5. Which function does an Aggregator Transformation perform?

A) Joins multiple datasets

B) Applies filter conditions

C) Performs calculations like SUM, AVG, MIN, MAX ✓

D) Splits data into multiple groups

6. What is a key characteristic of a Joiner Transformation?

A) Joins data only from relational databases

B) Supports only inner joins

C) Allows joins between heterogeneous sources ✓

D) Only works with flat files

7. What is the purpose of Rank Transformation?

A) Assigning a unique sequence number

B) Filtering duplicate values

C) Sorting data without ranking it

D) Finding the top or bottom 'N' records ✓

8. How does Sequence Generator Transformation work?

A) Generates unique primary keys ✓

B) Assigns random values to rows

C) Filters duplicate records

D) Joins multiple sources

9. A Lookup Transformation is mainly used for:

A) Joining two data sources

B) Retrieving related data from another dataset ✓

C) Filtering records based on a condition

D) Aggregating data

10. Which type of Slowly Changing Dimension (SCD) updates historical data while keeping the old data?

A) Type 1

B) Type 2 ✓

C) Type 3

D) None of the above

11. In Informatica, which transformation allows dynamic filtering based on real-time conditions?

A) Expression Transformation

B) Router Transformation ✓

C) Filter Transformation

D) Aggregator Transformation

12. What is a key advantage of using Informatica's pushdown optimization?

A) Moves transformation logic to the database for improved performance ✓

B) Reduces memory usage but increases execution time

C) Only works with flat files

D) Works only on source systems

13. What is a primary function of an Update Strategy Transformation?

A) Filters duplicate data

B) Controls insert, update, delete operations in target tables ✓

C) Joins multiple data sources

D) Assigns sequence numbers

14. When does Informatica create an Index Cache?

A) While performing Aggregator transformations

B) During Lookup transformations ✓

C) While using Router transformations

D) Only in Joiner transformations

15. What is the purpose of a Normalizer Transformation?

A) Denormalizes the data

B) Converts multiple columns into multiple rows ✓

C) Joins multiple sources

D) Aggregates data

16. What is the difference between a Passive and an Active Transformation?

A) Passive transformations remove duplicate records

B) Active transformations modify the number of rows ✓

C) Passive transformations filter data

D) Active transformations cannot change row count

17. In Informatica, what is a Mapplet?

A) A reusable set of transformations

B) A mapping that cannot be used in multiple workflows ✗ (Correct Answer: A) A reusable set of transformations)

18. What does the Union Transformation do?

A) Combines data from multiple sources ✓

B) Sorts data in ascending order

C) Removes duplicate rows

D) Joins tables using a primary key

19. What does the Transaction Control Transformation do?

A) Controls commit and rollback operations ✓

B) Deletes unnecessary records

C) Filters out null values

D) Assigns a rank to records

20. What is the purpose of the Parameter File in Informatica?

- A) Stores variable values that change at runtime** ✓
- B) Controls the ETL job execution schedule
- C) Stores permanent transformation values
- D) Defines the mapping structure

Tableau MCQs - Answer Key

1. What is Tableau primarily used for?

- A) Database management
- B) Data visualization and business intelligence** ✓
- C) Data entry and form processing
- D) Web application development

✗ Your Answer: A

2. What happens when you join two tables in Tableau?

- A) The tables are merged with all columns ✗
- C) The tables are combined based on a common key** ✓

✗ Your Answer: A

3. Which chart type is best for showing trends over time?

- A) Pie Chart
- B) Bar Chart
- C) Line Chart** ✓
- D) Scatter Plot

✓ Your Answer: C

4. What are Dimensions in Tableau?

- A) Numerical fields used for calculations
- B) Fields that cannot be used for visualization
- C) Categorical fields used for slicing and dicing data** ✓
- D) Aggregated values

✓ Your Answer: C

5. What are Measures in Tableau?

- A) Categorical fields
- B) Text-based attributes
- C) Numerical fields that can be aggregated** ✓
- D) Fields used only in filters

☒ Your Answer: C

6. What does Level of Detail (LOD) expression in Tableau do?

- A) Controls the granularity of aggregations ☒
- B) Filters data at the worksheet level
- C) Only works with Calculated Fields
- D) Controls the layout of the dashboard

☒ Your Answer: A

7. Which type of calculation in Tableau allows you to manipulate row-level data?

- A) Table Calculation
- B) LOD Expression
- C) Row-Level Calculation ☒
- D) Aggregate Calculation

☒ Your Answer: D

8. What is a primary function of Tableau Extensions?

- A) To modify data sources
- B) To customize and extend Tableau functionalities ☒
- C) To perform database backups
- D) To create dashboards only

☒ Your Answer: B

9. Which of the following is NOT a function type in Tableau?

- A) String Functions
- B) Aggregation Functions
- C) Visualization Functions ☒
- D) Logical Functions

☒ Your Answer: B

10. Which type of join in Tableau returns only matching records from both tables?

- A) Inner Join ☒
- B) Left Join
- C) Right Join
- D) Full Outer Join

☒ Your Answer: A

11. Which feature in Tableau helps create interactive dashboards?

- A) Parameters
- B) Filters
- C) Actions** ✓
- D) Calculated Fields

✓ **Your Answer: C**

12. What is a Dual-Axis Chart in Tableau?

- A) A chart with two Y-axes for comparison** ✓
- B) A combination of bar and line charts
- C) A pie chart with two measures
- D) A chart only used in dashboards

✓ **Your Answer: A**

13. In Tableau, what does the 'Show Me' panel do?

- A) It suggests different chart types based on selected data** ✓
- B) It filters out unnecessary columns
- C) It creates calculated fields automatically
- D) It removes duplicate records

✓ **Your Answer: A**

14. What does the FIXED LOD expression do?

- A) Aggregates data at a fixed level** ✓
- B) Dynamically filters data
- C) Changes field types automatically
- D) Applies conditional formatting

✓ **Your Answer: A**

15. What does the WINDOW_SUM function do in Tableau?

- A) Calculates running total over a partition** ✓
- B) Aggregates data across sheets
- C) Works only on Dimensions
- D) Filters out duplicate values

✓ **Your Answer: A**

16. What is the main purpose of Data Blending in Tableau?

- A) To combine multiple data sources without modifying them** ✓
- B) To create new columns
- C) To merge tables permanently
- D) To perform row-wise calculations

✓ **Your Answer: A**

17. What is the function of Extracts in Tableau?

- A) To store a snapshot of the data for faster performance** ✓
- B) To remove null values
- C) To create permanent database backups
- D) To delete duplicate records

✓ **Your Answer: A**

18. What does the ATTR function in Tableau do?

- A) Returns the first value of a field
- B) Aggregates categorical data while preventing multiple values** ✓
- C) Creates a new calculated field
- D) Removes missing values

✓ **Your Answer: B**

19. What is the default file format of Tableau Workbooks?

- A) .twb** ✓
- B) .twbx
- C) .tds
- D) .tde

✓ **Your Answer: A**

20. What is the purpose of a Story in Tableau?

- A) To create animations
- B) To combine multiple dashboards into a presentation** ✓
- C) To add dynamic filters
- D) To automate data refresh

✓ **Your Answer: B**